A New Method for Kalman Filter Tuning

Aykut Görkem Gelen
Department of Electrical and Electronics Engineering
Erzincan Binali Yıldırım University
Erzincan, Turkey
aykut.gelen@erzincan.edu.tr

Ayten Atasoy
Department of Electrical and Electronics Engineering
Karadeniz Technical University
Trabzon, Turkey
ayten@ktu.edu.tr

***Abstract*—Kalman filter is a well-known estimator. For a target tracking scenario, the filter needs two input parameters. These parameters are called process noise covariance and measurement noise covariance. It is necessary to determine these parameters in order to design a consistent and high-performance target tracker. These parameters vary widely depending on the problem and measurement system. Therefore, there is no commonly accepted method to tune the filter. In this study, we examined the performance and consistency of Kalman filter through a simple target tracking scenario. As a result of the study, three metrics are defined. These metrics can be calculated using measurement data and estimation results only. In the study, it has been shown that Kalman filter can be tuned using this metric. It is not necessary to know the actual states to calculate the metric. This fact makes the method useful for practical applications.**

***Index Terms*—Kalman Filter, Target Tracking, Tuning, Consistency**

I. Introduction

Kalman filter is an estimator that is used in many areas from military target tracking applications to economic predictions. Some extended versions or Kalman filter based methods are used in various tracking algorithms with single, multiple, non-maneuvering and maneuvering targets. The Kalman filter, which is a natural tracker, performs well even when applied directly to simple target tracking scenarios. However, besides this natural capability and simplicity, a tough tuning process awaits filter designers.

The Kalman filter is a recursive filter. The filter takes a state vector of the target, a covariance matrix that belongs to the state, the process noise covariance matrix, Q, and the measurement noise covariance matrix, R. The filter advances the state vector by a fixed time interval. It calculates the next state and next state covariance matrix from previously available data. During these steps, the Q and R matrices are used as fixed inputs. Therefore, these two parameters need to be determined in order to design a suitable Kalman filter. Performance of the filter strictly depends on the proper selection of these parameters. The fact that the Q parameter depends on the process and it can not be modeled precisely makes the problem interesting.

It is necessary to analyze the filter in order to develop a method that will tune the filter. Such a method can be developed by trial and error method. As a first step, the following questions should be answered.

978-1-5386-6878-8/18/$31.00 ©2018 IEEE

- What is changing in the estimation results for different Q and R values?
- What is the effect of different Q and R values on the consistency of the filter?
- Do different Q and R values change the response time of the filter?

Finding answers to these questions and develop a method to tune the Kalman Filter is the motivation for this work.

In this study, a metric was proposed in order to tune the filter. The metric was designed to give the best parameter combination at the point where it is minimum. This metric is a heuristic measure. It is based on the idea of how the estimates of the consistent filter should be.

The paper is organized as follows. In section II, related works are given. In section III, the target tracking problem, Kalman filter basics, statistical analysis of the filter are given. The proposed method is presented in section IV. Sections V and VI cover simulation and results. The conclusion is given in section VII.

II. Related Work

There are many studies on tuning the Kalman filter. However, there is no widely accepted method for selection of Q and R parameters in the literature. Bayesian methods have been examined and compared with each other in a study where the latest methods for setting Q and R parameters are compared each other [1]. In a study examining the performance metrics of filters, the effects of Q and R parameters were investigated [2]. In this study, the Q and R parameters are examined only in terms of estimation error. In another study, two different costs were determined to tune filter parameters as like in this study [3]. The metrics defined in this study are mathematically derived using innovation and innovation covariance as well as consistency testing. All past works and present mathematical methods have been collected in a technical report [4]. In a paper involving filter performance, convergence and degradation studies were conducted to select the Q and R parameters in order to reduce the estimation error [5].

III. Kalman Filter as a Target Tracker

In this section, target tracking, Kalman filter and performance measures are given.

A. Target Tracking Problem

Tracking an object is the estimation of interested parameters by using measurements. In most cases, these parameters may not be directly measurable. In addition, there are always uncertainties in the measurements due to signal noise produced by environmental sources. At this point, mathematical models for process and measurement systems are used to estimate parameters. The system can be modeled approximately, even if inputs are not exactly known. As a consequence, the problem is how to estimate the interested features of the target using the information in the measurements available.

Since the measurements are taken at regular intervals, the problem requires a discrete time recursive solution. Therefore, the target can be expressed mathematically as a discrete-time dynamic system. In order to derive a model for the motion of the target, it is sufficient to write kinematic equations. A general discrete-time dynamic system can be express as in (1) and (2). In these equations, the state vector and measurements are represented by x_k and y_k respectively. The state transition matrix and measurement matrix are expressed by F and H. The process noise and measurement noise are denoted by v_k and w_k.

$$x_{k+1} = Fx_k + v_k \tag{1}$$

$$y_k = Hx_k + w_k \tag{2}$$

Many different dynamic models for target tracking have been developed [6]. These models are based on the kinematic equations of Newton. Kinematic equations are given in (3) and (4). Position, velocity, acceleration and time step denoted by p, v, a and ΔT respectively.

$$p_{k+1} = p_k + v_k\Delta T + a_k\Delta T^2 \tag{3}$$

$$v_{k+1} = v_k + a\Delta T \tag{4}$$

As a consequence, the target tracking problem is to estimate the state vector at distinct times, using the measurements and prior knowledge available up to related instant.

B. Kalman Filter

The Kalman filter is a recursive Bayesian filter. Using the Bayesian approach, the filter tries to find posterior probability by taking advantage of the prior knowledge and collection of measurements. The filter operates under the following assumptions [7].

- System and measurement models are linear.
- The process and measurement noise components are white, zero mean and normally distributed with covariances Q and R respectively.
- The initial state, i.e. a priori probability, is a normally distributed random variable with mean x_0 and covariance P_0.

The mean and covariance are sufficient to represent a random variable with the normal distribution. Hence, it can be considered that the means and covariances at the input and output of the Kalman filter, represent the prior and posterior probability distributions respectively.

Kalman filter needs a mathematical process model in order to predict the next state. This model is formed by kinematic equations. Since the actual state of the target can never be known, this model roughly describes the motion of the target. The Kalman filter uses both the model predicted state and measurements. One cycle of the Kalman filter is given in Algorithm 1.

Algorithm 1 One cycle of Kalman Filter

INPUT: Initial state and state covariance, (x_k, P_k), Covariance matrices, (Q, R), measurement y_k
OUTPUT: Estimated state and state covariance, $(\hat{x}_{k+1}, P_{k+1})$

1: Calculate predicted state, x_p, and predicted state covariance, P_p as $x_p = Fx_k, P_p = FP_kF^T + Q$
2: Calculate predicted measurement, y_p as $yp = Hx_p$
3: Calculate innovation covariance S_k as $S_k = HPpH^T + R$
4: Calculate filter gain, K as $K = P_pH^TS_k^-1$
5: Calculate estimated state and state covariance as $\hat{x}_{k+1} = x_p + K(y_k - y_p)$ and $P_{k+1} = P_p - KS_kK^T$

Since the filter is recursive, the state and state covariance calculated at the end of one cycle is used as the initial value for the next step.

C. Performance and Consistency

In the first cycle, Kalman filter needs for an artificial prior probability. Therefore, an initial state and initial state covariance must be defined to start the filter. This is referred to the problem of filter initiation in the literature [8]. In this study, the two-point initialization method, which is one of the simplest methods, was used. Filter initiation does not affect the overall performance of the filter. However, it affects the convergence time. For this reason, the filter initialization issue was not examined in this study. The other input parameters are covariance matrices Q and R. The values of these matrices should be determined intuitively because noise covariances in a real application cannot be known in advance.

The consistency in Kalman filter is defined as the coherence between estimated parameters and assumptions in section III-B [7]. Two parameters were defined in order to test consistency. The first parameter is the *Normalized Estimation Error Squared* (NEES). The second parameter is *Normalized Innovation Squared* (NIS). These parameters are given as (5) and (6)

$$\epsilon(k) = \tilde{x}(k)^T P(k)^{-1} \epsilon(k) = \tilde{x}(k) \tag{5}$$

$$\epsilon_v(k) = v(k)^T S(k)^{-1} v(k) \tag{6}$$

IV. Proposed Method

The purpose of this study was developing a method to find optimum values of the input parameters Q and R. At this point, it is assumed that there should be an optimal combination of Q and R parameters that make the filter consistent and makes the estimation error low. The noise affecting the measurements and the disturbances affecting the target motion is not exactly known. Therefore, it is quite difficult to set the values of these parameters. If the values of these parameters are set close to actual values, the filter will be converged and consistent. Otherwise, divergence will occur.

When a Kalman filter is designed for a real-life scenario, the actual values of state variables and the measurement noise are unknown. However, the filter designer has measurements, estimated states and covariances. At this point, the known data can be used for developing a method for Kalman filter tuning. Such a method is expected to determine the values of the matrices Q and R by using measurements , estimated states and covariances.

Many metrics which are measures of the performance and consistency of the filter can be defined. If such metrics can be developed, the problem reduces to a minimization or maximization problem. Three different metrics have been used in this study. These metrics are listed below.

1) **Combined Position Error**
This property indicates the distance between the i^{th} measurement y^i and the predicted measurement, y_p^i. In the Kalman filter, if the Q is too large, the Kalman gain will give more weight to the measurements. Therefore, the estimation results will be closer to the measurements. Since this metric is calculated from the distance between the measurement and the estimation results, the metric value is expected to decrease as Q grows. The calculation of Combined Position Error is given as Algorithm 2.

Algorithm 2 Algorithm to calculate Combined Position Error

INPUT: Collection of estimated state vectors (X), Measurement Model Matrix (H), the collection of measurements, (y)
OUTPUT: Median of combined position error, (P_1)

for all estimates $\hat{x}^i$ in set X **do**
    find the position difference vector D^i between measurement y^i and predicted measurement y_p^i
    calculate combined difference D_c^i by finding magnitude of the vector D^i
end for
calculate P_1 as the median of D_c

2) **One Hop Deviation**
This feature indicates the standard deviation of the differences of the estimated positions between time steps. Since the Kalman filter uses a single mathematical model, the differences between the estimation results should be similar. The calculation of One Hop Deviation is given as Algorithm 3.

Algorithm 3 Algorithm to calculate One Hop Deviation

INPUT: Collection of estimated state vectors (X), Measurement Model Matrix (H)
OUTPUT: Deviation of position differences between time steps, (P_2)

for all estimates $\hat{x}^i$ in set X except last one **do**
    find the position difference vector D^i between y_p^{i+1} and y_p^i
    calculate combined difference D_c^i by finding magnitude of the vector D^i
end for
for all combined difference D_c^i except last one **do**
    form difference vector M by calculating $D_c^{i+1} - D_c^i$
end for
calculate P_2 as standard deviation of M

3) **Filter Response** This property shows the response of the filter to the change in the measurements. If the R value is large, the Kalman filter will assume that the measurements have too many noise. Therefore, the filter will give more weight to the model and will not react to the changes in the measurements. This reality is used when calculating this metric. The calculation of Filter Response is given as Algorithm 4.

Algorithm 4 Algorithm to calculate Filter Response

INPUT: Collection of estimated state vectors, (X)
OUTPUT: Deviation of position differences between time steps, $(p2)$

form D_X vector by calculating differences between adjacent elements of y_p
form D_Y vector by calculating differences between adjacent elements of y
form combined positions D_{XC} and D_{YC} by calculating magnitudes of D_X and D_Y respectively
form M_{XC} and M_{YC} by calculating group of n elements of D_{XC} and D_{YC} respectively
calculate difference vector K by subtracting M_{XC} and M_{YC}
calculate P_3 as standard deviation of K

At this point, Metric 1 and Metric 3 should be combined to determine Q values, Metric 2 and Metric 3 should be combined to find R values. Combination of increasing and decreasing metrics is done to find an optimum point. The maximum operator was chosen as combination operator. This selection ensures that the optimum point is minimum. Logically, optimal Q and R values should make all the metrics minimum.

These three metrics are combined using the method given in Algorithm 5. Calculated metrics need to be formed as a matrix so that rows and columns show different Q and R values respectively. In this algorithm, scaling was done on Metric 1 and Metric 2 to cross metrics at an optimum point.

Algorithm 5 Algorithm to combine metrics

INPUT: Metrics, (P_1, P_2, P_3)

OUTPUT: The optimal values of covariance matrices, (Q, R)

find mean of rows of P_1 as Cr_1 and divide by 2.75 for scaling
find mean of rows of P_3 as Cr_3
calculate the difference between maximum and minimum values of Cr_1 as DF_1
calculate the shift for Cr_3 in order to find the crossing point by using fitted polynomial $P_Q(DF_1)$
re-calculate Cr_3 as $2mean(Cr_3) - Cr_3 + shift$
find Q by $min(max(Cr_1, Cr_3))$
find mean of columns of P_2 as Cc_2 and divide by 2.75 for scaling
find mean of rows of P_3 as Cc_3
calculate the difference between maximum and minimum values of Cc_2 as DF_2
calculate the shift for Cc_3 in order to find the crossing point by using fitted polynomial $P_R(DF_2)$
re-calculate Cc_3 as $CC_3 + shift$
find R by $min(max(Cc_2, Cc_3))$

In the Algorithm 5, the differences between the maximum and minimum values of Metric 1 and Metric 2 are used to determine the shift amount for Metric 3. For this purpose, the actual amount of shift is calculated for the critical points determined. Using the obtained data, a 7th order polynomial is fitted which calculates the shift amounts using the difference between max and min differences.

V. Simulation

In this study, constant velocity target tracking scenarios were created to test the developed metric. The system in equations (1) and (2) were used to generate actual states and measurement data.

For this study, it is assumed that target tracking is performed in Cartesian coordinate system and measurements are two-dimensional Cartesian positions. State vector for constant velocity model is given in (7). In this vector, coordinates are denoted by ξ, η. Velocities are denoted by first derivatives respectively.

$$x_{cv} = \begin{bmatrix} \xi & \dot{\xi} & \eta & \dot{\eta} \end{bmatrix}^T \tag{7}$$

The constant velocity model in state transition matrix and measurement matrix form are given in (8), (9).

$$F_{CV} = \begin{bmatrix} 1 & \Delta T & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & \Delta T \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{8}$$

$$H_{CV} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \tag{9}$$

As the process noise, it is assumed that there are small changes in the acceleration for both motion models. The process noise was calculated as in (10).

$$v = G \begin{bmatrix} a_\xi \\ a_\eta \end{bmatrix} \tag{10}$$

$$G_{CV} = \begin{bmatrix} \frac{\Delta T^2}{2} & 0 \\ \Delta T & 0 \\ 0 & \frac{\Delta T^2}{2} \\ 0 & \Delta T \end{bmatrix} \tag{11}$$

The accelerations in (10) are calculated as in (12) where randn(2,1) is 2×1 normally distributed random number vector.

$$\begin{bmatrix} a_\xi \\ a_\eta \end{bmatrix} = \sigma_Q \times randn(2,1) \tag{12}$$

As the measurement noise, normally distributed random numbers with σ_R standard deviation was used as in (12).

The Q and R matrices are given in (13) and (14) where 0_2 and I_2 are 2×2 zeros and identity matrix. The Q_{11} and Q_{22} are equal and given in (15).

$$Q = c_Q \begin{bmatrix} Q_{11} & 0_2 \\ 0_2 & Q_{22} \end{bmatrix} \tag{13}$$

$$R = c_R I_2 \tag{14}$$

$$Q_{11} = \begin{bmatrix} \frac{\Delta T^3}{3} & \frac{\Delta T^2}{2} \\ \frac{\Delta T^2}{2} & \Delta T \end{bmatrix} \tag{15}$$

Using this system, 100 actual state vectors and 100 measurements are generated. The estimation results and metrics are calculated and recorded by using these data. As a procedure, metrics for different Q and R values were recorded for scenarios where real Q and R values are known. The Q values ranged from 0.1 to 4 in steps of 0.25. The R values ranged from 30 to 70 in steps of 2.5. All these operations are repeated 1000 times to make a Monte Carlo simulation. Metrics are combined and c_Q and c_R values are calculated by Algorithm 5.

VI. Results and Discussion

In this section, the results of a target tracking simulation with actual Q and R values of 1 and 50, were presented in order to introduce the proposed method and show the metrics. The measurement data and estimation results were recorded and metrics were calculated. Metric 1 and Metric 3 are plotted together and given as Fig. 1. Metric 1 values were found to be higher than metric 3 as seen in this figure. As expected, the metric 1 is decreasing with increasing Q values. As can be seen from the Fig. 1, there are significant changes in Q values in Metric 1. However, the values are always decreasing. It needs to be compared with a metric with increasing Q in order to find an optimum point. This can happen if the Metric 3 is scaled and shifted.

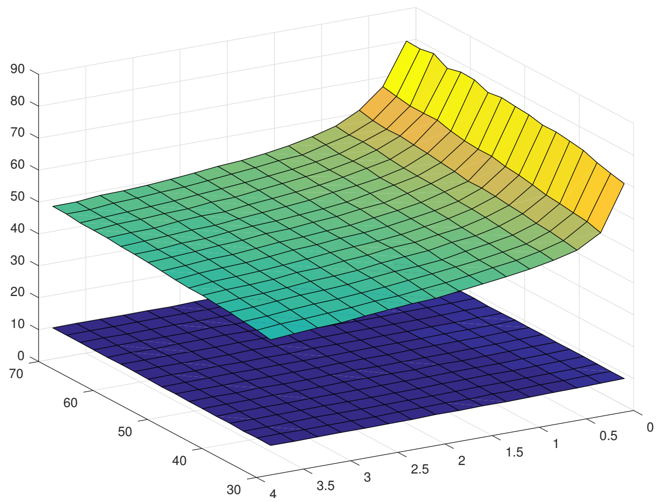

Fig. 1. 3D plot of Metric 1 and Metric 3

Metric 2 and Metric 3 are plotted together and are given in Fig. 2. In this figure, it can be seen that the Metric 2 values are decreasing and the Metric 3 values are increasing along the R axis. Additionally, Metric 2 values increase as Q values increase. In order to determine optimal R, Metric 3 needs to be scaled and shifted.

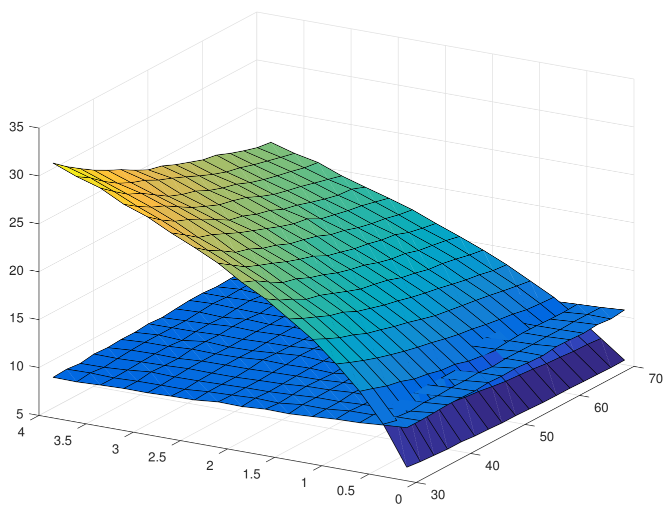

Fig. 2. 3D plot of Metric 2 and Metric 3

In order to show metrics better, the metrics are rearranged into two separate 2D drawings for Q and R. Arranged metrics are given in Fig. 3. In order to avoid more than one curve for each metric in the figure, mean of columns for Q and mean of rows for R are plotted.

In figure 3, the metric 1 was divided by 2.75 in order to scale it. This number was chosen by trial and error to move metrics closer. Metric 3 was scaled as given in Algorithm 5. This scaling converts the decreasing metric 3 values to the increasing values with the same average. The results are given in Fig. 4.

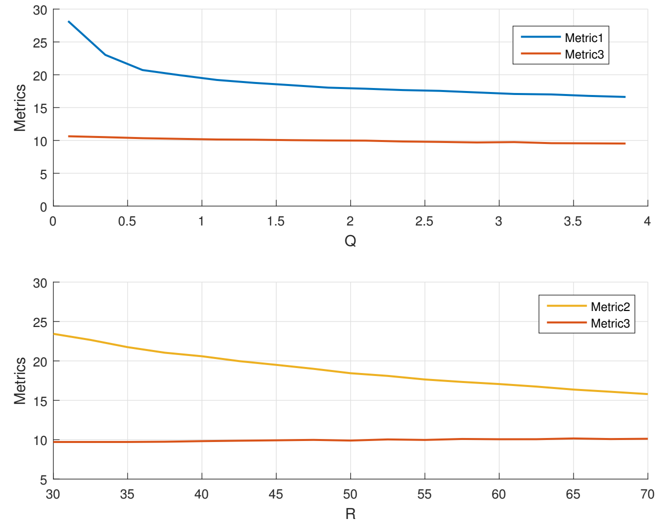

Fig. 3. Mean of the metrics in 2D

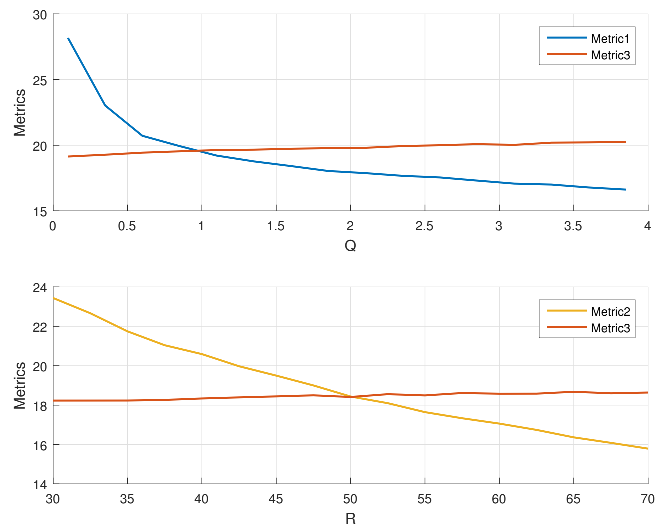

Fig. 4. Scaled and Shifted Metrics

As can be seen from the figure, there are intersections at the values of 1 and 50 which are optimum values for Q and R. Using this method, the values of the Q and R parameters can be found for the applications where the measurements and the estimation results are recorded. In order to calculate the values used for scaling, all values of the metric must be known. This is a disadvantage for the proposed method in terms of the calculation time. In addition, in order to apply the method, the range of Q and R parameters must be known roughly.

In this study, different combinations of Q and R values were generated and the metrics were calculated. The recommended method has been tried for all these data and optimum Q and R values were found. Calculated metrics were used to find polynomial coefficients that calculate the amount of shift. Additional data was also generated in order to test the method.

VII. Conclusions

In this study, a method to find the Q and R values to make the Kalman filter consistent is proposed. The values obtained by calculating the three different metrics defined. These metrics are compared with each other and the optimal solution is determined. The use of measurement data and estimation results to calculate metrics allow the proposed method to be used in real applications. In order to apply the procedure used to adjust metrics, all values of the metrics must be calculated at certain intervals. This situation extends the duration of the simulation, but as a result the optimal values of Q and R are determined.

The feasibility of the method has been tested with constant velocity target tracking simulations. As a result, the proposed method found optimal Q and R values.

References

[1] P. Matisko and V. Havlena, "Noise covariances estimation for kalman filter tuning," *IFAC Proceedings Volumes*, vol. 43, no. 10, pp. 31–36, 2010.

[2] K. Akçay, "Performance metrics for fundamental estimation filters," Master's thesis, Middle East Technical University, 2005.

[3] M. Saha, B. Goswami, and R. Ghosh, "Two novel costs for determining the tuning parameters of the kalman filter,"

[4] S. M. M, N. Naik, R. M. O. Gemson, and M. R. Ananthasayanam, "Introduction to the kalman filter and tuning its statistics for near optimal estimates and cramer rao bound,"

[5] D.-J. Jwo and T.-S. Cho, "A practical note on evaluating kalman filter performance optimality and degradation," *Applied Mathematics and Computation*, vol. 193, pp. 482–505, nov 2007.

[6] X. R. Li and V. P. Jilkov, "Survey of maneuvering target tracking. part i. dynamic models," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 39, pp. 1333–1364, Oct. 2003.

[7] Y. Bar-Shalom, T. Kirubarajan, and X.-R. Li, *Estimation with Applications to Tracking and Navigation*. New York, NY, USA: John Wiley & Sons, Inc., 2002.

[8] M. Mallick and B. L. Scala, "Comparison of single-point and two-point difference track initiation algorithms using position measurements," *Acta Automatica Sinica*, vol. 34, pp. 258–265, mar 2008.